

Proposed Solution

Date	15/07/2026
Team ID	SWTID-2026-3095
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Project proposal and specifications report.

Project Overview

Objective: Automate credit card approval decisions by predicting an applicant's credit risk from application and repayment history data, cutting manual review time and reducing bad-debt exposure.

Scope: Merging applicant and credit history records, feature engineering, encoding categorical variables, correcting for severe class imbalance, training Logistic Regression, Random Forest, Decision Tree and XGBoost classifiers, tuning the best model, and serving predictions through a Flask web app.

Problem Statement

Manual credit card underwriting is slow and inconsistent, and often misses applicants likely to default. Card issuers need an automated, data-driven screening tool that flags high-risk applicants early using historic repayment behavior, reducing bad debt while approving creditworthy customers quickly.

Resource Requirements

Resource Type	Description / Allocation	Specification
Hardware	Developer workstation	Standard CPU, 8GB+ RAM, 10GB free disk (GPU not required)
Software	Development stack	Python 3.11, VS Code, Git, Flask web server
ML Libraries	Data science tools	scikit-learn, XGBoost, pandas, numpy, joblib, scipy
Data Source	Historic applicant & credit records	Kaggle: application_record.csv (438,557 rows), credit_record.csv (1,048,575 rows)